

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY

9647/03

Paper 3 Free Response

18 September 2015

2 hours

Candidates answer on separate paper.

Additional Materials: Writing Papers
Data Booklet
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **four** questions.
Begin each question on a fresh sheet of paper.

You are advised to show all working in calculations.
You are reminded of the need for good English and clear presentation in your answers.
You are reminded of the need for good handwriting.
Your final answers should be in 3 significant figures.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part question.

This document consists of **11** printed pages and **1** blank page.



Answer any **four** questions.

- 1** Phosphorus forms three chlorides, P_2Cl_2 , PCl_3 and PCl_5 , of which the latter two are the most common. Both PCl_3 and PCl_5 are valuable chemical intermediates in many processes.

(a) By considering bonding and structure, explain why PCl_3 is a gas while PCl_5 is a solid at room temperature and pressure. [2]

(b) (i) Phosphorus oxychloride, $POCl_3$, is an intermediate compound formed during the reaction between PCl_5 and water.

Draw a dot-and-cross diagram to show the bonding in $POCl_3$ and state the shape of the molecule. [2]

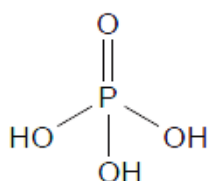
(ii) Write the equation of reaction of PCl_5 with water to form phosphorus oxychloride. [1]

(iii) Reactions of covalent chlorides with water can be rationalised as step-wise replacement of $-Cl$ with $-OH$.

There are 3 steps in the mechanism for the formation of phosphorus oxychloride from phosphorus pentachloride. Given step 1, complete the mechanism by writing the equations for steps 2 and 3.

Step 1: $PCl_5 + H_2O \longrightarrow PCl_4OH + HCl$ [2]

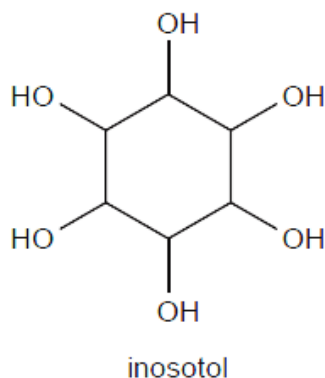
(c) PCl_5 also reacts with water to form phosphoric acid, H_3PO_4 . The structure of phosphoric acid is shown below.



(i) Phosphoric acid may dimerise to produce diphosphoric acid, $H_4P_2O_7$ and water. The reaction involves the condensation of an $-OH$ from each H_3PO_4 molecule to create an oxygen bridge between the two phosphoric acid units.

Draw the structure of diphosphoric acid. [1]

- (ii) Recent research from a group led by Nobel-prize winning chemist Jean-Marie Lehn has shown that a compound containing three cyclic diphosphates can improve the body's capacity for exercise. The compound is a derivative of inositol which is shown below.

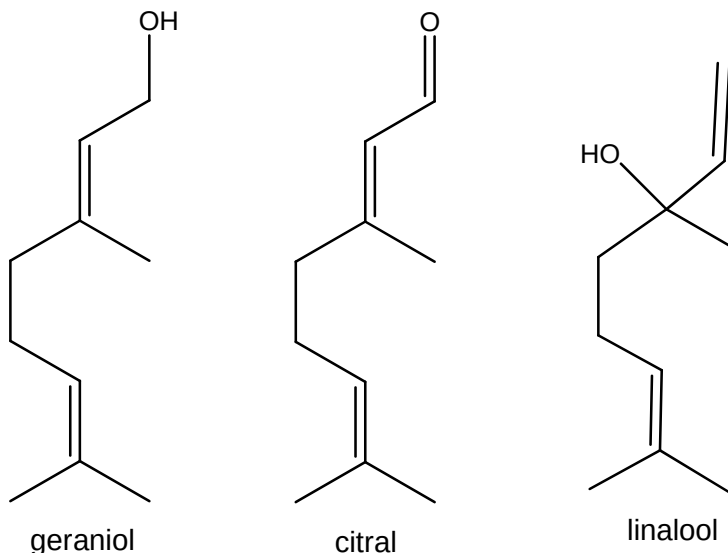


In the new compound each pair of adjacent oxygen atoms from the inositol molecule is part of a cyclic diphosphate ester. The overall charge of the inositol-cyclic-phosphate species is -6 .

Suggest a structure for the compound.

[1]

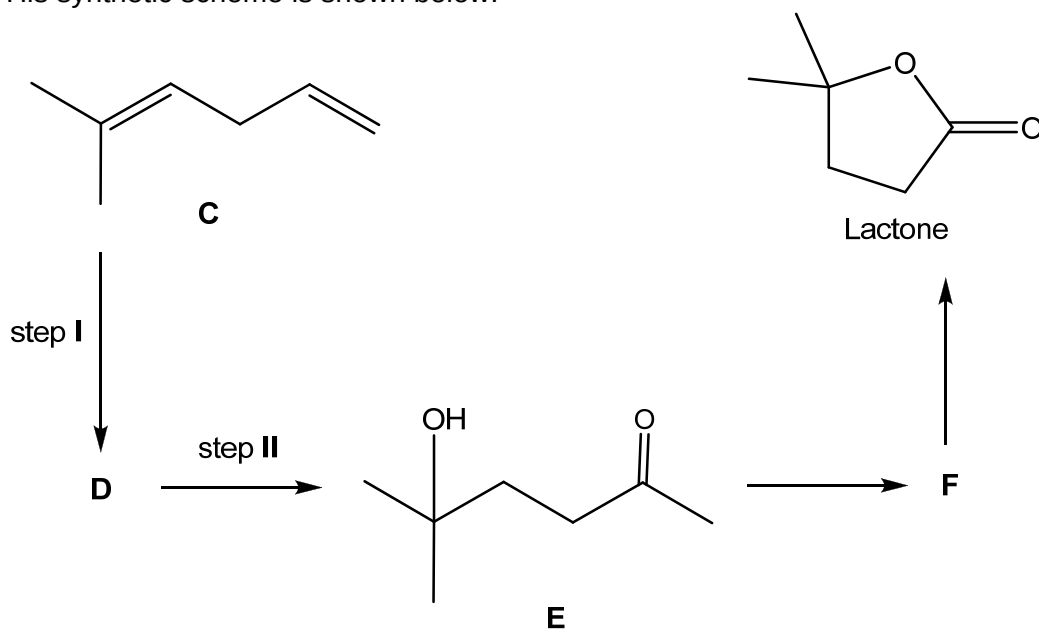
- (d) Describe the reactions, if any, of the oxides P_4O_{10} and SiO_2 with water. Include the approximate pH value of any resulting solutions, and write equations for any reactions that occur.
- (e) Geraniol, citral and linalool are important flavours in citrus fruits. These compounds are often used in the food and perfume industries.



Suggest **two** simple chemical tests that could be used to distinguish the three compounds from each other. You should state what you would observe for **each** compound in **each** test.

[4]

- (f) A laboratory scientist attempted to synthesise a lactone starting from hydrocarbon **C**. His synthetic scheme is shown below.



- (i) Suggest the reagents and conditions required for steps I and II. [2]
- (ii) Give the structural formulae of compounds **D** and **F**. [2]

[Total: 20]

- 2 Aluminium is a silvery-white, soft metal which is the third most abundant element in the Earth's crust. Aluminium has a relatively low density, and is resistant to corrosion. Compounds of aluminium have also been used commercially for many purposes.

(a) $\text{Al}(\text{OH})_3$ is a sparingly soluble aluminium compound with a K_{sp} value of 3×10^{-33} that can be used as a feedstock for manufacturing other aluminium compounds, such as aluminium chloride, zeolites, sodium aluminate.

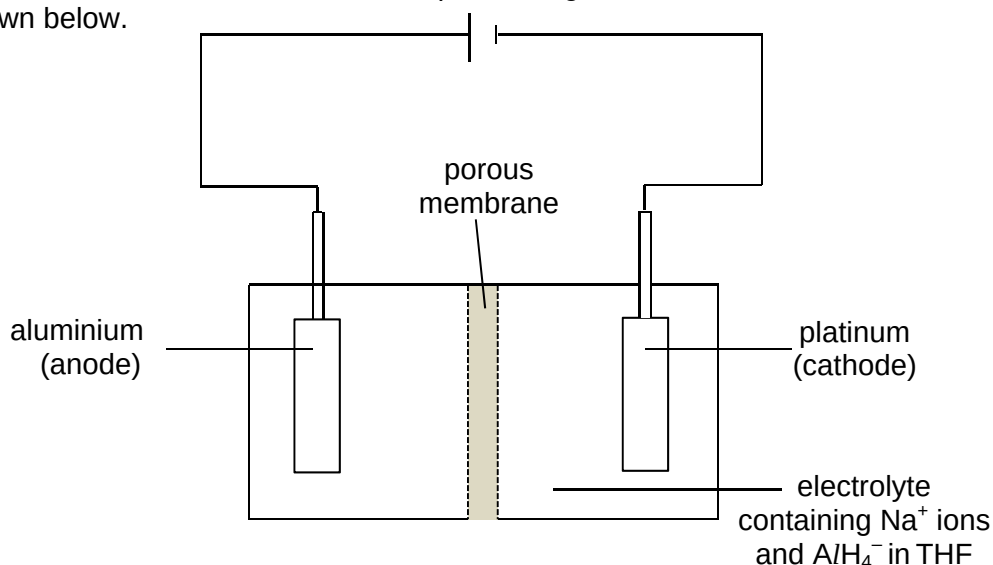
(i) Write an expression for the K_{sp} of $\text{Al}(\text{OH})_3$, stating the units. [2]

(ii) Use the data above to determine the solubility of $\text{Al}(\text{OH})_3$ in water. [1]

(iii) Predict, with a reason, how the solubility of $\text{Al}(\text{OH})_3$ will change when it is dissolved in 0.10 mol dm^{-3} of $\text{Al}(\text{NO}_3)_3$. [2]

(iv) $\text{Al}(\text{OH})_3$ is amphoteric in nature. Write balanced equations to illustrate this. [2]

(b) Alane, AlH_3 , can also be produced under chemically inert environment from the electrolysis of a solution containing Na^+ ions and AlH_4^- ions in an organic solvent such as anhydrous tetrahydrofuran (THF). Aluminium is used as the anode, and platinum is used as the cathode. A simplified diagram of the electrochemical set-up is shown below.



Alane and hydrogen gas are produced at the anode, while sodium metal is collected at the cathode. THF is not involved in the reaction. A current of 2A is passed through the set-up for 40 minutes.

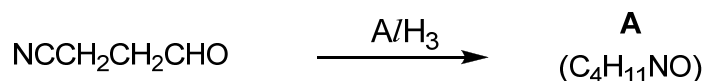
(i) Write half-equations for the reactions at the anode and cathode respectively. [2]

(ii) A side-reaction occurred at the anode. It was observed that the anode became thinner over time. Given that alane is the only product in this side-reaction, suggest an ion-electron equation to account for this observation. [1]

(iii) Based on the relevant equation in (i), calculate the amount of sodium collected from the experiment above. [1]

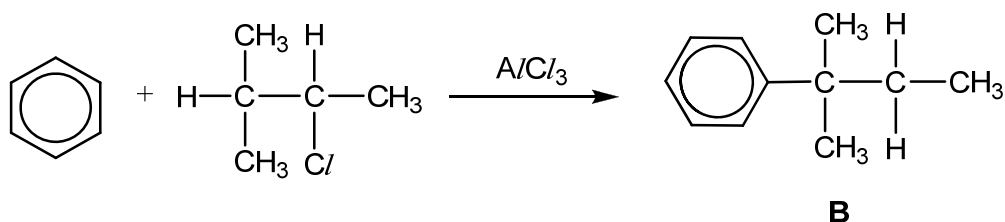
(iv) At room temperature and pressure, 473 cm^3 of hydrogen gas is formed at the anode. Based on your equations in (i) and (ii), calculate the amount of alane produced from the side-reaction. [2]

- (c) AlH_3 can be used as a reagent in organic reactions. Compound **A** can be synthesised using AlH_3 in the following reaction.



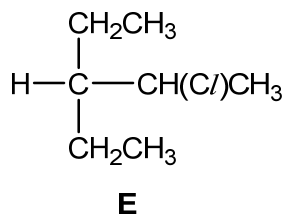
A shows negative result with Brady's reagent and reacts with sodium metal to produce a gas. Suggest the structure of **A**, and explain your answer. [3]

- (d) AlCl_3 can be used as a catalyst for the reaction of 2-chloro-3-methylbutane with benzene to form **B**.



The reaction proceeds through the formation of a carbocation **1**. As carbocation **1** is not very stable, a carbocation rearrangement occurs. In the rearrangement, a hydride shift is involved to form a more stable carbocation **2**.

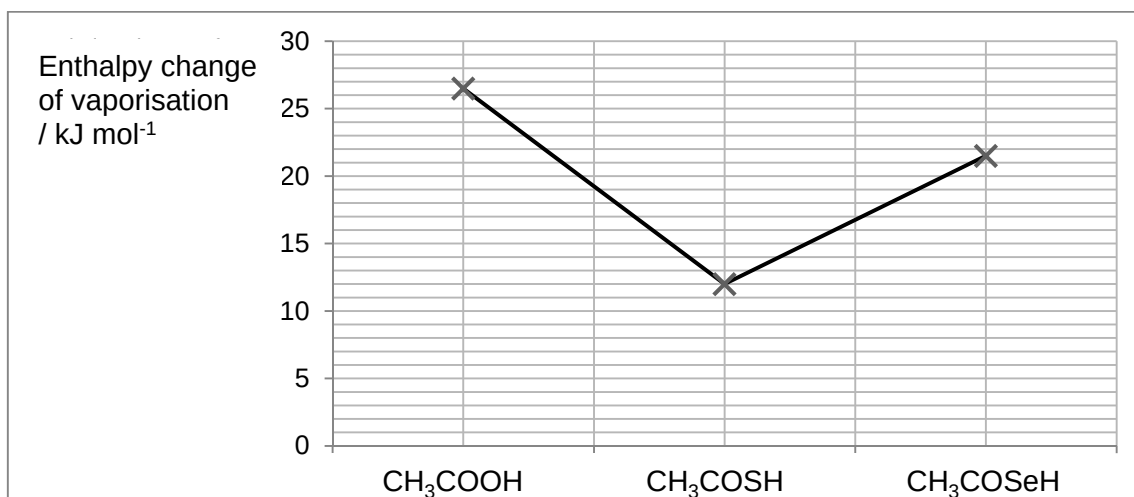
- (i) Suggest the structures of both carbocations. Label your answers clearly, and explain why carbocation **2** is more stable than carbocation **1**. [3]
- (ii) Suggest the structure of product formed if compound **E** is used instead of 2-chloro-3-methylbutane for the above reaction.



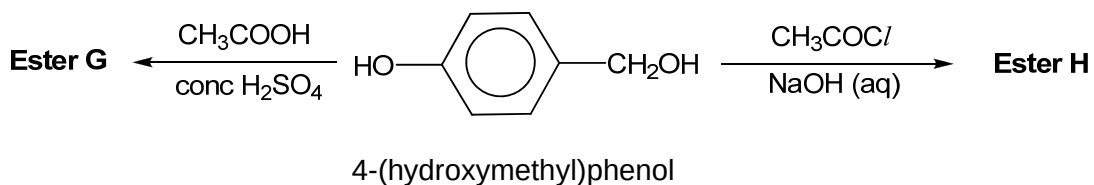
[1]

[Total: 20]

- 3 (a) The graph below shows the enthalpy change of vaporisation of CH_3COOH , CH_3COSH and CH_3COSeH .



- (i) Explain the trend above. [3]
- (ii) Ethanoic acid is found to have an apparent molar mass of 120 g mol^{-1} when it is heated to just above its boiling point. Draw a labelled diagram to illustrate this phenomenon. [1]
- (iii) By making use of the above graph and your answer to (i), deduce the enthalpy change of hydrogen bond formation. [2]
- (iv) Hence, or otherwise, determine the enthalpy change of dimerisation of ethanoic acid. [1]
- (v) Given that the entropy change of dimerisation is $-130 \text{ kJ mol}^{-1} \text{ K}^{-1}$, explain why dimerisation is spontaneous at low temperature under the same pressure. [2]
- (b) Ethanoic acid and its derivative can react with 4-(hydroxymethyl)phenol to form two different esters under suitable conditions.



- (i) Draw the structural formulae of ester **G** and **H**. [2]
- (ii) Briefly explain why an alkaline medium is necessary for the formation of ester **H**. [1]

- (c) Collagen, a component of bone and connective tissue, is the most abundant protein in vertebrates. It exists as water-insoluble fibres of great strength.

Five of the most common amino acids in the collagen obtained from human skin are listed below.

Amino acid	Formula of side chain (R in $\text{RCH}(\text{NH}_2)\text{COOH}$)
glycine (gly)	$-\text{H}$
alanine (ala)	$-\text{CH}_3$
glutamic acid (glu)	$-\text{CH}_2\text{CH}_2\text{COOH}$
serine (ser)	$-\text{CH}_2\text{OH}$
lysine (lys)	$-(\text{CH}_2)_4\text{NH}_2$

- (i) There are three pK_a values associated with glutamic acid: 2.1, 4.1 and 9.5.

Sketch the pH-volume curve obtained when 10 cm^3 of 0.1 mol dm^{-3} protonated glutamic acid is titrated against 0.1 mol dm^{-3} $\text{NaOH}(\text{aq})$ until 40 cm^3 of $\text{NaOH}(\text{aq})$ is added. Indicate the various key points on the curve. [3]

- (ii) Draw the two major forms of glutamic acid present at pH 2.1. [2]

- (d) (i) Human haemoglobin undergoes denaturation in a strongly basic medium. With respect to the globular protein subunit, explain why denaturation takes place. [2]

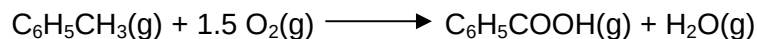
- (ii) Hemochromatosis is an autosomal recessive genetic disease that results in deregulated iron absorption in specific organs such as the liver and heart. Liver enzymes acts as a catalyst and an increase in amount of the enzymes will aggravate the condition of a patient diagnosed with hemochromatosis.

Sketch the Boltzmann distribution curve to show how the presence of the liver enzyme aggravates the condition. [1]

[Total: 20]

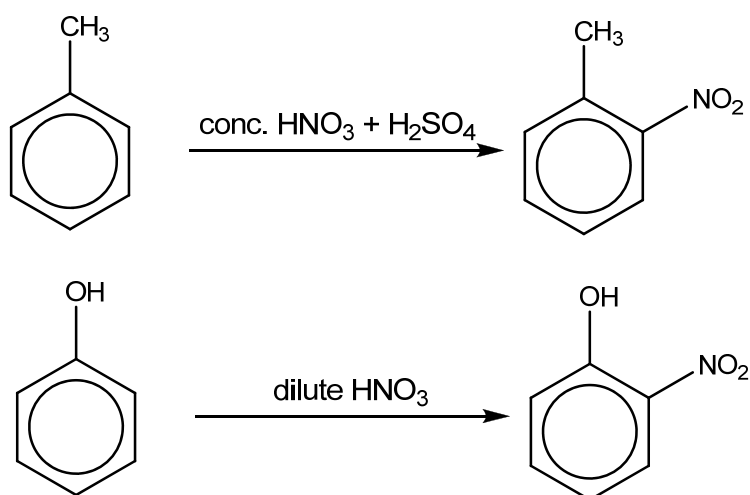
- 4 (a) Methylbenzene is a colourless aromatic hydrocarbon that is obtained from petroleum by distillation. It is used as a solvent and as a starting material for the synthesis of many organic compounds, including dyes and explosives.

It can be oxidised to benzoic acid using oxygen over a suitable catalyst according to the following equation.

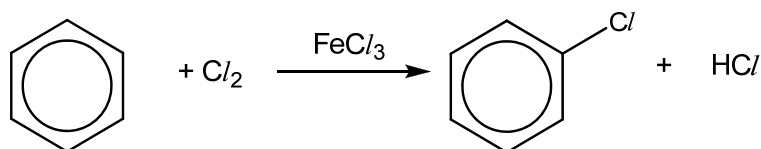


- (i) Use bond energy values from the *Data Booklet* to calculate the enthalpy change for the reaction. [2]
- (ii) Using your answer from (i), construct a reaction pathway diagram for this reaction. [2]

- (b) Both methylbenzene and phenol can react with nitric acid, but under different conditions.



- (i) Explain why different conditions are needed for these two reactions. [2]
- (ii) Suggest and explain how the acidity of 2-nitrophenol might compare with that of phenol. [2]
- (c) Iron(III) chloride is used as a catalyst in electrophilic substitution reactions. The chlorination of benzene is represented by the following overall equation.

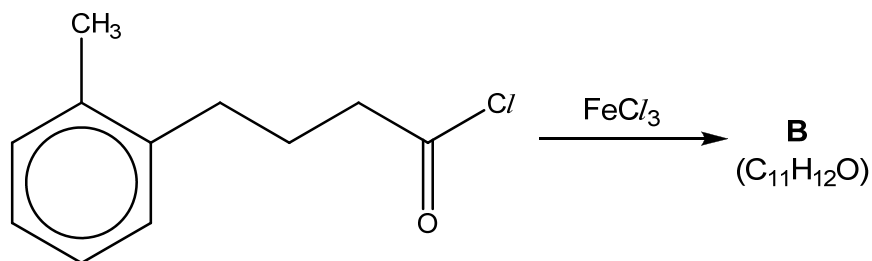


The reaction occurs in several steps.

- The first step is the reaction between Cl_2 and FeCl_3 .
- The benzene ring is then attacked by the Cl^+ cation in the second step.

FeCl_3 reacts in a similar way with acyl chlorides, producing a carbocation that can then attack a benzene ring.

Describe the mechanism for the following reaction:



[2]

- (d) A $0.150 \text{ mol dm}^{-3}$ solution of iron(III) chloride was found to have a pH of 1.52.
- (i) Write an expression for the acid dissociation constant, K_a of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$. [1]
 - (ii) Calculate the K_a of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$. [1]
 - (iii) Explain why the pH of a solution of iron(II) chloride is higher than that of a solution of iron(III) chloride of the same concentration. [1]
- (e) Use of the Data Booklet is relevant to this question.

When aqueous iron(II) chloride is boiled with an excess of potassium cyanide, a yellow solution **D** is obtained. Acidification of **D**, followed by the passage of chlorine gas through the solution, gives an orange solution **E**. Iron(II) chloride is found to catalyse the reaction between I^- and $\text{S}_2\text{O}_8^{2-}$ whereas **E** does not affect the rate of this reaction.

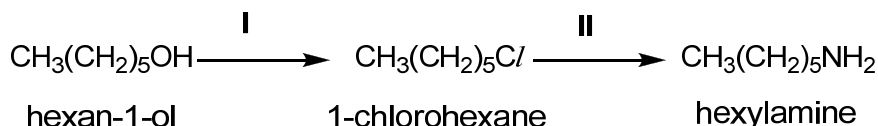
Solution **E** can also be formed by adding aqueous potassium cyanide to brown precipitate **B**. Precipitate **B** is formed by adding excess NH_3 to aqueous iron(II) chloride and letting it stand in air.

- (i) Explain why a solution of iron(II) chloride appear green. [2]
- (ii) Identify solution **D** and suggest the type of reaction that has occurred for the formation of **D** from iron(II) chloride solution. [2]
- (iii) Identify precipitate **B**. [1]
- (iv) Identify solution **E**. [1]
- (v) Explain why **E** does not affect the rate of reaction between I^- and $\text{S}_2\text{O}_8^{2-}$. [1]

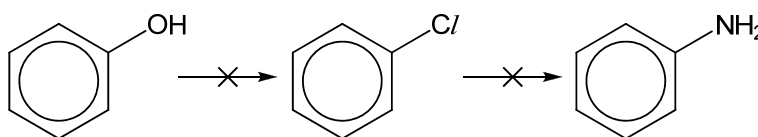
[Total: 20]

5 The halogens and their compounds are useful laboratory reagents.

- (a) Describe how the reactions of chloride and iodide ions with concentrated H_2SO_4 differ. You should include suitable observations and balanced equations for any reactions that occur. [4]
- (b) Phenol and chlorobenzene are less reactive towards certain reagents than similar non-aromatic compounds. Thus hexan-1-ol can be converted into hexylamine by the following two reactions:



whereas neither of the following two reactions takes place.



- (i) Suggest reagents and conditions for reactions I and II. [2]
- (ii) What type of reaction is reaction II? [1]
- (iii) Suggest a reason why chlorobenzene is much less reactive than 1-chlorohexane. [2]
- (iv) State and explain how the rate of reaction II changes when 1-chlorohexane is replaced by 1-iodohexane. [2]
- (c) The compounds **P** and **Q** are isomers with the molecular formula $\text{C}_9\text{H}_8\text{O}_2$.

P liberates carbon dioxide with sodium hydrogencarbonate but **Q** does not. **Q** reacts with an ammoniacal solution of silver nitrate followed by acidification to give **R** and a grey precipitate.

Upon oxidation with hot acidified potassium manganate(VII), 1 mole of **P** gives 2 moles of carbon dioxide and 1 mole of benzoic acid while 1 mole of **Q** gives 1 mole of carbon dioxide and 1 mole of a compound, $\text{C}_8\text{H}_6\text{O}_5$.

P decolourises aqueous bromine in the dark to form **S**, $\text{C}_9\text{H}_9\text{O}_3\text{Br}$ while **Q** reacts with aqueous bromine to form a white precipitate **T**, $\text{C}_9\text{H}_6\text{O}_3\text{Br}_4$. When compound **S** is heated with acidified sodium dichromate(VII) solution, the solution remains orange.

Deduce the structures of all the lettered compounds (**P** to **T**) and explain the chemistry of the reactions described. [9]

[Total: 20]

